

is even larger: 79.6% vs 29.6% (50pp). This occurs despite the frontier model being the teacher that generated the training data.

3. The improvement over base is large. Across tasks, the CoT-augmented model improves over the base model by 5.9pp (UHCS) to 67.6pp (granulometry). On three of four tasks (excluding UHCS, where the base model’s 60.8% is inflated by majority class bias), the method achieves $2.5\times$ to $6.6\times$ the base model’s accuracy.

5.2 Granulometry Results

Granulometry is a 9-class problem with two classification axes. Table 5 breaks down accuracy per axis.

The grading axis benefits most from CoT distillation (+7.4pp vs +1.9pp for size). This is expected: grading requires understanding whether gaps between stones are empty (coarse), partially filled (medium), or completely filled (fine). The CoT descriptions explicitly teach these gap-pattern distinctions, which is harder to learn from JSON-only supervision.

5.3 Weld Defect Results

The weld defect task produces the strongest headline result: the 3B model beats GPT-4.1 few-shot by 10.8 percentage points. Table 6 shows per-class accuracy.

Cracks remain the hardest class. All baseline models score 0% on cracks in zero-shot mode. With just 6 training images of cracks, the augmented model reaches 50%. The frontier model reaches only 30% even with few-shot examples. This suggests the fine-tuned small model learns crack-specific features more effectively than the frontier model’s in-context learning.

On this task, Direct LoRA slightly outperforms CoT-augmented on the cracks class (58% vs 50%), while CoT-augmented is better overall. This per-class variation suggests that the two approaches learn different decision boundaries.

5.4 Steel Surface Results

Steel surface defects present the most challenging scenario for our method. GPT-4.1 few-shot achieves 91.1% on this task, well above our fine-tuned model’s 66.7%. Table 7 shows per-class results. The gap exists because GPT-4.1’s few-shot setting includes reference images for each class, providing direct visual comparisons that are highly effective for these 200×200 grayscale images.

Inclusion is consistently the hardest class (32% augmented, 38% direct). Inclusions appear as dark elongated streaks visually similar to scratches in grayscale images. The augmented approach shows improvement on scratches (63% to 85%) and patches (52% to 75%), demonstrating that contrastive reasoning helps where visual features are more distinct. Overall, the augmented approach still outperforms Direct LoRA (66.7% vs 63.1%).

5.5 UHCS Microstructure Results

UHCS shows the smallest improvement from CoT distillation (+0.9pp). The base model already achieves 60.8% zero-shot (inflated by spheroidite majority class bias), and GPT-4.1 few-shot reaches 71.7%. Our CoT-augmented model achieves 66.7%, approaching but not exceeding the frontier model. Table 8 shows per-class results.